• Math refresher

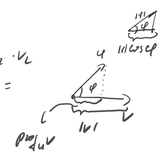
• dot product

$u = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$ $v = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$

$\langle u, v \rangle = u_1 v_1 + u_2 \cdot v_2$

$= \text{proj}_u v \cdot |u| = \langle u, v \rangle$

$= |v| |u| \cos\varphi =$



• Matrix-vector multiplication

$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} - z_1 - \\ - z_2 - \end{bmatrix} = \begin{bmatrix} | & | \\ c_1 & c_2 \\ | & | \end{bmatrix}$

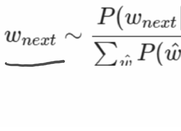
$u = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$

row picture

$Au = \begin{bmatrix} au_1 + bu_2 \\ cu_1 + du_2 \end{bmatrix} = \begin{bmatrix} \langle z_1, u \rangle \\ \langle z_2, u \rangle \end{bmatrix} = u_1 \begin{bmatrix} | \\ c_1 \\ | \end{bmatrix} + u_2 \begin{bmatrix} | \\ c_2 \\ | \end{bmatrix}$

column picture

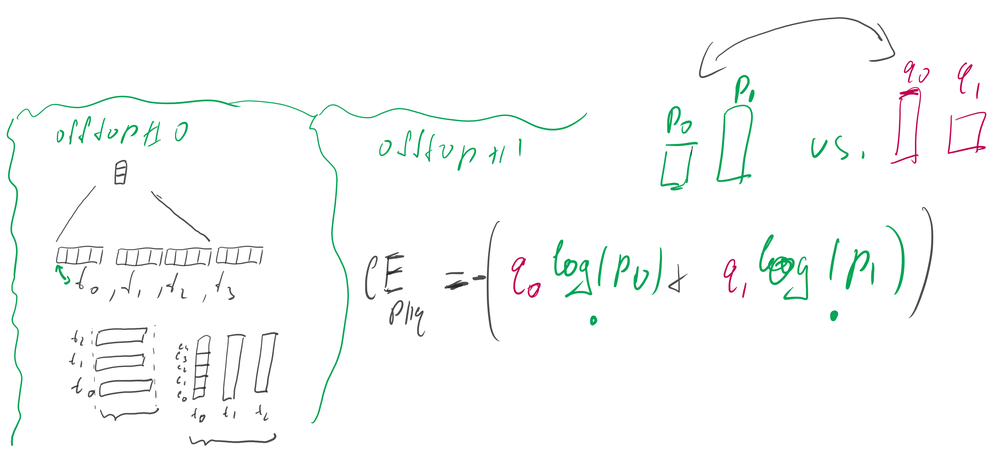
$v = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} u_1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ u_2 \end{bmatrix}$



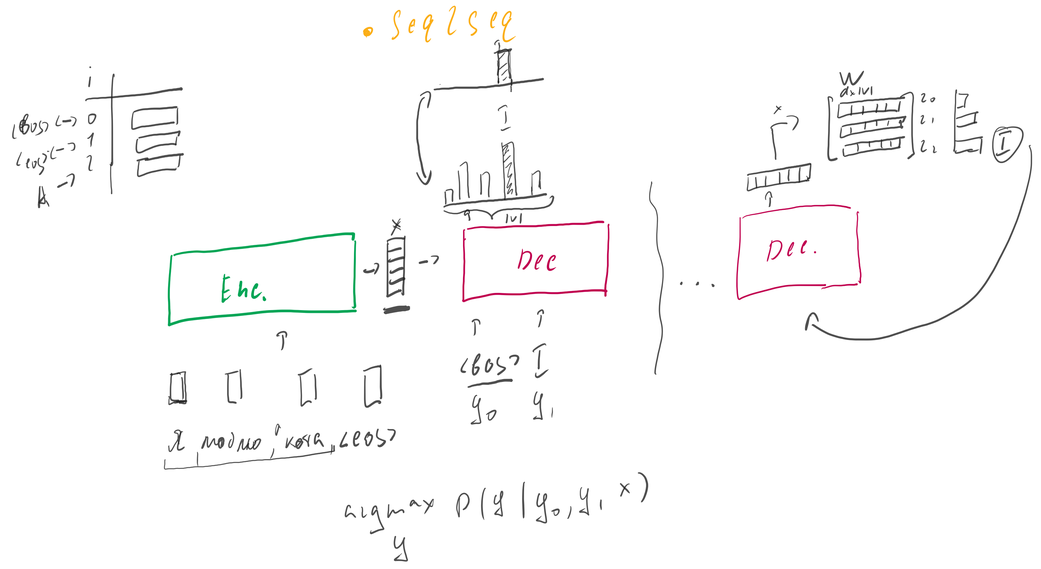
$u = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} | & | \\ e_1 & e_2 \\ | & | \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$

$A = \begin{bmatrix} - \\ - \end{bmatrix}$ $B = \begin{bmatrix} | & | \\ c_1 & c_2 \\ | & | \end{bmatrix}$

$AB = \begin{bmatrix} | & | \\ Ac_1 & Ac_2 \\ | & | \end{bmatrix}$

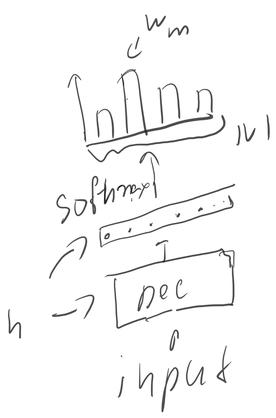


• Seq2Seq



$\text{argmax}_y P(y \mid y_0, y_1, x)$

• Temperature



$w_{next} \sim \frac{P(w_{next}|X)^{1/\tau}}{\sum_{\hat{w}} P(\hat{w}|X)^{1/\tau}}$

$P(w_m) = \frac{exp(w_m)^{1/T}}{\sum_{i=0}^{|V|} exp(w_i)^{1/T}}$

• Log-exp trick:

$a^k = \exp(\log(a^k))$;

$\exp(\log(\exp(w_m)^{1/T})) =$

$= \exp(\frac{1}{T} \log(\exp(w_m))) =$

$= \exp(\frac{1}{T} w)$

$T \uparrow \rightarrow$

$T \downarrow \rightarrow$